Databases

From system of record to system of reason: The rise of the AI-native database

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The database must transform into an active "system of reason," capturing the autonomous agent's explainable thought process to govern trust and control in the AI-native business era.

Editor's note: This article originally ran in InfoWorld.

For decades, the database has been the silent partner of commerce: a trusted, passive ledger. It was the system of record, the immutable vault that ensured every action had an equal, auditable reaction. This model underwrote the entire global economy. But that era of predictable, human-initiated transactions is over.

We are entering the agentic era. A new class of autonomous agents — systems that perceive, reason, act, and learn — are becoming the primary drivers of business operations. These systems do more than just execute prescribed workflows. They generate emergent, intelligent behavior. This creates a profound new challenge for leadership: in a business increasingly run by autonomous systems, how do you ensure trust, control, and auditability? Where is the handshake in a system that thinks for itself?

The answer is not to constrain the agents, but to evolve the environment in which they operate. The

intelligent platform that serves as the agent's conscience. Beyond recording *what* an agent did, it must provide an immutable, explainable 'chain of thought' for *why* it did it. This is the dawn of the AI-native database.

Three mandates for leadership

- **Evolve your database from passive ledger to active reasoning engine.** Your data platform needs to be more than a repository. It must become an active participant in informing, guiding, and enabling autonomous action.
- **Build your durable AI advantage with an enterprise knowledge graph.** Sustainable differentiation will not come from the AI model alone, but from the comprehensiveness of your proprietary data, structured as a graph of interconnected entities that powers sophisticated reasoning.
- **Establish an 'AgentOps' framework for high-velocity deployment.** The primary bottleneck in delivering AI value is the human workflow. The platform that wins is the one that provides the most productive and reliable path from concept to production-grade autonomous system.

reason to ensure control and trust in autonomous operations.

- Advantage requires a data moat: a proprietary enterprise knowledge graph and unified HTAP+V foundation.
- To ship AI value fast, use AgentOps, the integrated framework for reliable deployment.

Phase 1: Perception — giving agents high-fidelity senses

An agent that cannot perceive its environment with clarity and in real-time is a liability. The Home Depot built their 'Magic Apron' agent to move beyond simple search to provide expert 24/7 guidance, pulling from real-time inventory and project data to give customers tailored recommendations. This level of intelligent action requires a unified perception layer that provides a complete, real-time view of the business. The foundational step is to engineer an AI-native architecture that converges previously siloed data workloads.

Unifying real-time senses with HTAP+V

now, while analytical warehouses show what happened in the past. Agents operating on this divided architecture are perpetually looking in the rearview mirror. The solution is a converged architecture: Hybrid Transactional/Analytical Processing (HTAP). Google has engineered this capability by deeply integrating its systems, allowing BigQuery to directly query live transactional data from Spanner andAlloyDB without impacting production performance.

The agentic era requires a new sense: intuition. This means adding a third critical workload — vector processing — to create a new paradigm: **HTAP+V**. The 'V' enables semantic understanding, allowing an agent to grasp intent and meaning. This technology recognizes a customer asking 'where is my stuff?' has the same intent as someone reporting a 'delivery problem.' Recognizing this, Google has integrated high-performance vector capabilities across its entire database portfolio, enabling powerful hybrid queries that fuse semantic search with traditional business data.

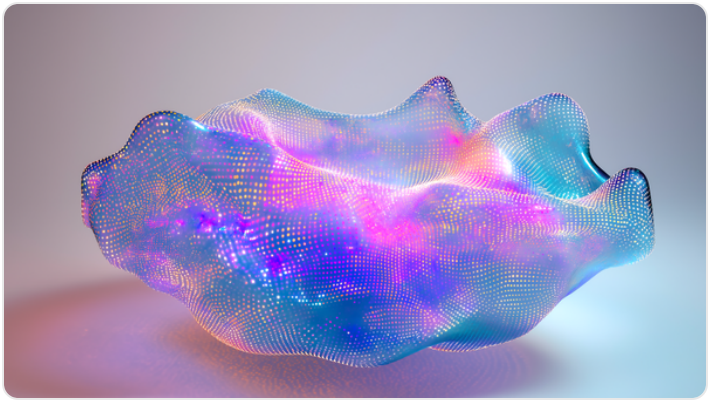
Teaching agents to see the whole picture

An enterprise's most valuable insights often sit in unstructured data: contracts, product photos, support call transcripts. Agents need fluency across all these formats. This requires a platform that treats multimodal data not as a storage problem, but as a core computational element. This is precisely the future BigQuery was built for, with innovations that

3, which models the complex interactions of molecules from a massive multimodal knowledge base, is a profound demonstration of this power. If this architecture can unlock the secrets of biology, it can unlock new value in your business.

A control plane for perception

Perfect perception without governance creates risk. Machine-speed decisions now outpace traditional, manual governance. The solution is to build agents that operate within a universe governed by rules. This requires transforming the data catalog from a passive map into a real-time, AI-aware control plane. This is the role of Dataplex, which defines security policies, lineage, and classifications once and enforces them universally — ensuring an agent's perception is not only sharp, but foundationally compliant by design.



The ROI of AI

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Phase 2: Cognition — architecting memory and reasoning

Perception alone isn't enough. Agents also need understanding. This requires a sophisticated cognitive architecture for memory and reasoning. Consider a financial services agent that uncovers complex fraud rings in minutes by reasoning across millions of transactions, accounts, and user behaviors. This demands a data platform that is an active component of the agent's thought process.

Engineering a multi-tiered memory

Agents require two types of memory:

1. **Short-term memory:** A low-latency 'scratchpad' for immediate tasks, requiring absolute consistency. Spanner, with its global consistency, is precisely engineered for this role and is used by platforms like Character.ai to manage agent workflow data.
2. **Long-term memory:** The agent's accumulated knowledge and experience. BigQuery, with its

allowing agents to retrieve the precise "needle" of information from a petabyte-scale haystack.

The moat: Connective reasoning with knowledge graphs

Memory alone doesn't enable reasoning. Standard Retrieval-Augmented Generation (RAG) is like giving an agent a library card — it can find facts, but it can't connect the ideas. GraphRAG represents the next evolution, enabling agents to traverse different information sources like a researcher connecting ideas. As vector search becomes commoditized, the enterprise knowledge graph becomes the true, durable moat. This is the future Google is engineering with native graph capabilities (GQL) in its databases, a vision validated by DeepMind research on Implicit-to-Explicit (I2E) Reasoning showing that agents become exponentially better at complex problem-solving when they can first build and query a knowledge graph.

Phase 3: Action — building an operational framework for trust

In the agentic era, velocity creates competitive advantage: the speed at which you transform ideas into a production-grade autonomous process.

phase is about building the high-velocity 'assembly line' to govern an agent's actions reliably and safely.

The agent's conscience: Embedded intelligence and explainability

Trust requires transparent reasoning. This starts with bringing AI directly to the data. Today, platforms like BigQuery ML and AlloyDB AI do this by embedding inference capabilities directly within the database via a simple SQL call. These systems are further augmented by emerging Knowledge Graph capabilities that capture the intelligence about data and business context. This transforms the database into the agent's conscience.

But inference alone is not enough. The next frontier of trust is being pioneered by DeepMind through advanced capabilities that are becoming part of the platform. This includes a new generation of Explainable AI (XAI) features, informed by DeepMind's work on data citation, which allows users to trace a generated output back to its source. Before agents act in the physical world, they need safe practice environments. DeepMind's research with models like the SIMA agent and 'Generative Physical Models' for robotics demonstrates the mission-critical importance of training and validating agents in diverse simulations — a capability being integrated to de-risk autonomous operations.

Once you establish trust, speed becomes the priority. Human workflows create the bottleneck. A new operational discipline – AgentOps' – is required to manage the lifecycle of autonomous systems. Major retailers like Gap Inc. are building their technology roadmaps around this principle, using Vertex AI platform to accelerate their e-commerce strategy and bring AI to life across their business. The platform's Vertex AI Agent Builder provides a comprehensive ecosystem from a code-first Python toolkit (ADK) to a fully managed, serverless runtime (Agent Engine), all while leveraging BigQuery for the massive-scale log analytics required to monitor and debug agent behavior. This integrated toolchain is what solves the "last mile" problem, collapsing the development and deployment lifecycle.

Moving forward in the AI-native era

The transition to the agentic era requires architectural and strategic changes:

1. **Unify the foundation (Perception):** Create an AI-native architecture built on converged HTAP+V workloads, integrating platforms like AlloyDB, Spanner, and BigQuery under a single governance plane.

beyond simple chatbots. Prioritize a tiered memory architecture and invest in a proprietary enterprise knowledge graph as your central competitive moat.

3. **Master the last mile (Action):** Invest in a comprehensive AgentOps practice centered on an integrated platform like Vertex AI, which is what separates failed experiments from transformative business value.

This integrated stack provides what the agentic era demands: agents that can perceive accurately, reason deeply, and act with trust at machine speed.

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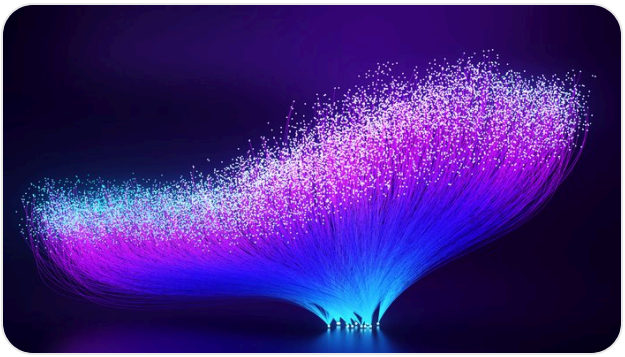

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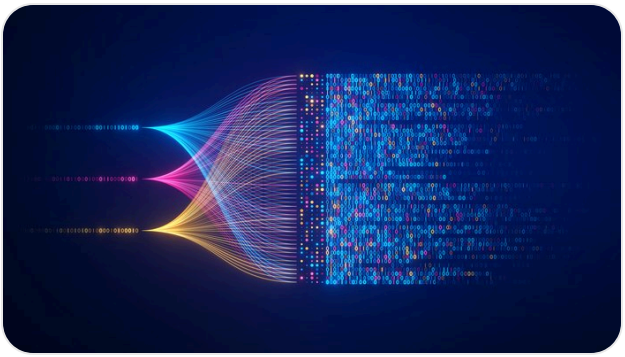

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